

Module 3 Contribution Journal

ITAI 1371 - Machine Learning

Neural Crew

Viviana Zamora

For the Module 3 lab, I completed the Machine Learning Workflow and Types of Learning notebook in Google Colab. I worked through the notebook section by section, ran the code cells, checked the outputs, and completed the written analysis in the reflection cell. The lab helped me connect the definitions of supervised, unsupervised, and reinforcement learning to actual examples instead of only memorizing the terms.

I worked with the Wine dataset and reviewed its structure, including the number of samples, features, classes, and missing values. I followed the workflow for preparing data, splitting it into training and testing sets, training classification models, and evaluating their accuracy. I compared Logistic Regression with a Decision Tree and identified Logistic Regression as the better-performing model in my results. I also considered how changing the selected features could affect the model and wrote that I would test other features to see whether the accuracy improves.

My individual analysis included a real-world application related to real estate. I described predicting home prices from information such as location, square footage, number of bedrooms and bathrooms, and the age of the property. I identified this as supervised learning because the model would learn from homes with known sale prices. This example made the assignment more meaningful to me because it connected machine learning to a field I already understand.

In addition to completing my own notebook, I served as the team lead for Neural Crew. I created and maintained the GitHub repository, collected the files submitted by team members, reviewed the repository contents, and merged the pull request that was submitted. I organized the team's materials so that the professor could access the final work from one GitHub link. I also checked the file names and deliverables and prepared the team's journals to be compiled into the required contribution and reflection files.

My main contribution was therefore both technical and organizational. I completed the lab work and analysis while also handling the final repository and submission responsibilities. Managing the team files showed me why clear naming, version control, and one final working copy are important in a group project. It also gave me practical experience using GitHub as a collaboration tool instead of only as a place to store my own work.

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Dustin Mastrota

For the Module 3 lab, I contributed by completing the Machine Learning Workflow and Types of Learning notebook along with my group. Since our group members are all completing the same lab and working through the same material, my contribution was mainly making sure I understood the notebook, completed the required sections correctly, and was able to compare my work with the group if anyone had questions or needed help checking something.

I worked through the notebook in Google Colab and ran the code cells as I went instead of just running everything at once. I am still learning Python and machine learning, so I tried to slow down and look at what each section was doing. I paid attention to the outputs and the way the code used the dataset to train and test the model. This helped me better understand the workflow instead of only focusing on getting the assignment finished.

One part of the lab I completed was the section about the different types of machine learning. I answered the questions that asked whether certain examples were supervised learning, unsupervised learning, or reinforcement learning. I reviewed the differences between them before answering because it can be easy to mix them up at first. Supervised learning uses labeled data to make predictions, unsupervised learning looks for patterns in data without labels, and reinforcement learning involves learning through rewards and consequences. Completing that part helped me understand why different types of machine learning are used for different problems.

I also worked with the Wine dataset and the classification model in the notebook. In one section, we were able to choose three features and see how they affected the model's accuracy. I looked at the results and compared the accuracy from the selected features to the accuracy from the original model. This showed me that the features used in a machine-learning model can make a difference in how well it performs. A model does not always improve just because changes are made. Sometimes a different feature selection can lower the accuracy, which is why testing and comparing results is important.

Another part of my contribution was completing the written reflection and analysis questions in the notebook. I explained the three types of machine learning in my own words and wrote about the model results. I also thought about why Logistic Regression may have performed well for the dataset and what could possibly be changed or tested in the future to improve the model.

For my real-world machine-learning example, I focused on the cinema and movie-theater industry. I thought about how theaters or movie companies could use data from ticket sales over

time to see what types of movies people in a particular city are most likely to watch. For example, a theater could look at information such as movie genre, rating, release date, ticket sales, showtimes, season of the year, and location. The data could show whether people in one city tend to watch more horror movies, action movies, family movies, comedies, or R-rated movies compared to people in another city.

Machine learning could use that information to help theaters decide which kinds of movies may be more successful in certain locations. For example, if ticket-sale data shows that horror movies and R-rated thrillers perform especially well in one city during October, a theater could use that information when deciding how many showtimes to schedule for similar movies. It could also help movie studios and theaters with advertising, promotions, and deciding which movies should have a wider release in certain areas. I thought this was a good example because it shows how machine learning can use past customer data to make predictions about what people may want to see in the future.

I also made sure my notebook was completed and saved correctly. I checked my assessment answers, reviewed the written reflection, and made sure the notebook was ready to be shared with the group. Since we are all doing the same work, it is still helpful to compare that everyone completed the requirements and that nobody missed an important section. This also gives us a chance to discuss parts of the assignment that may have been confusing.

Overall, I feel that I contributed by carefully completing the Module 3 notebook, checking my answers, reviewing the machine-learning concepts, working with the Wine classification model, and completing the written reflection. Even though our group members all complete the same lab, I believe my contribution is making sure that I do my part thoroughly and understand the material well enough to participate in group discussions and help check the final work. I am still new to machine learning and Python, but I feel more comfortable than I did at the beginning of the course.

Module 3: Contribution Journal

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Neural Crew .

Name: Funsho Orogun .

For this lab, I first reviewed the three main types of machine learning: supervised, unsupervised, and reinforcement learning. This was important to set the stage for what kind of problem I'd be solving. I knew this lab would focus on supervised learning, so I was looking forward to diving into that.

I needed to get my environment ready. That's why I imported all the necessary libraries like pandas for data manipulation, numpy for numerical operations, matplotlib and seaborn for visualizations, and various tools from Sklearn for machine learning. Having everything at hand makes the workflow smooth.

Then, I loaded the Wine dataset. I chose this dataset because it's a classic for classification problems, and it's built right into scikit-learn, which makes it super easy to access. I immediately converted it into a pandas DataFrame because I find it much easier to work with than raw NumPy arrays, and I added columns for the wine class names to make it more readable. I checked its shape and class names to get a quick overview.

I did some initial exploratory data analysis (EDA). I looked at the overall shape, the number of features, and especially the distribution of the wine classes. It's good to see that there are no missing values, which means less cleaning for me! I also plotted the class distribution to see if the classes were balanced (they're fairly well-balanced, which is good for training) and a correlation heatmap of the first few features to understand if any features were strongly related to each other. This was the core of my lab, applying the 6-step ML workflow:

1. **Data Preparation:** For simplicity, I decided to pick just four features (alcohol, malic_acid, ash, and alcalinity_of_ash) for my initial model. I separated these features (X) from the target variable (wine_class, or y).
2. **Data Splitting:** This is a crucial step! I split my data into training (80%) and testing (20%) sets. I used random_state=42 to ensure my results were reproducible, and stratify=y to make sure the class proportions were maintained in both the training and testing sets. This prevents the model from just memorizing the training data and gives me a way to evaluate its performance on unseen data.
3. **Model Training:** I chose to train two different classification models: Logistic Regression and a Decision Tree. Logistic Regression is a good baseline linear model, and a Decision Tree is a non-linear model. I set random_state=42 for both to

ensure reproducibility and limited the `max_depth` of the Decision Tree to 3 to prevent it from becoming too complex and overfitting early on.

4. **Model Evaluation:** After training, I needed to see how well my models performed. I used the held-out test set to make predictions and then calculated the accuracy score for both models. I also generated a detailed `classification_report` which gives me precision, recall, and F1-score for each class, which is much more informative than just accuracy, especially if classes were imbalanced. Based on this, Logistic Regression turned out to be the better performer with my chosen features.
5. **Model Interpretation:** To understand how the best model was making its predictions and where it was making mistakes, I visualized the confusion matrix for the Logistic Regression model. The diagonal tells me correct predictions, and the off-diagonal cells show misclassifications. It's a great way to see which classes are being confused with each other.

I also took a moment to review different data types like continuous, discrete, nominal, ordinal, text, and boolean. This section reminded me that the type of data dictates how you preprocess it and what algorithms are most suitable.

Hands-On Practice

For hands-on practice, I tried selecting a different set of features (`alcohol`, `color_intensity`, `proline`) to see if I could improve the model's accuracy. It's an iterative process, and trying different features is a fundamental part of optimizing performance. My new combination didn't beat the original, but it showed me the impact of feature choice!

I believe that I succeeded by identifying ML types, which was a good check of my foundational knowledge. Finally, the reflection section The most important thing I learned throughout this lab was the importance of the entire ML workflow, from data understanding to iterative evaluation.

The most challenging part was truly understanding evaluation metrics beyond just accuracy, like the nuances of the confusion matrix. Moving forward, in the future, I would love to explore how to handle imbalanced datasets, systematic feature selection, and effective hyperparameter tuning without overfitting.

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Patrick Ray

For the Module 3 lab, my main contribution was completing my part of the Machine Learning Workflow and Types of Learning notebook. I worked through the notebook in Google Colab and ran the cells so I could see the results as I went through the assignment thoroughly looking for mistakes. Since I am still pretty new to machine learning and Python, I tried not to just run everything without looking at it. I spent time looking at the outputs and trying to understand the layout and what the code was actually doing.

One part I worked on was identifying different types of machine learning. I completed the assessment where I had to decide if different situations were supervised, unsupervised, or reinforcement learning. I got 5 out of 5 corrections on that section. This helped me contribute to the assignment because I was able to show that I understood the differences between the three types instead of just running the code.

I also worked with the Wine dataset and the classification model. One of the parts of the lab allowed me to choose three features and see how well the model performed using them. I selected malic acid, flavonoids, and hues. When I ran the model with those features, I got about 83.3% accuracy. The original model had about 88.9% accuracy. At first, I was a little confused because I thought I was supposed to get a better score, but after looking at the results I understood that changing the features can actually make the model perform better or worse. This was one of the more useful parts of the lab for me because I could actually see how a change I made affected the results.

Another contribution I made was completing the reflection and analysis section inside the notebook. I explained supervised, unsupervised, and reinforcement learning in my own words. I also wrote about why I thought Logistic Regression performed better, what I could try next to improve the model, and how machine learning could be used in a real world situation. For my real world example, I used transportation and self-driving vehicles because I think this is an interesting way machine learning can be used. I wrote about how a vehicle could use things such as camera images, traffic information, road conditions, speeds, and locations to help make driving decisions.

I also made sure the notebook was completed and saved correctly. I ran the assessment and checked that my answers were correct, and I reviewed the reflection section before finishing it. After completing my work, I downloaded the notebook and uploaded my completed Module 3 notebook to my GitHub repository. I made sure the file was named correctly on GitHub so that

my work could be shared with the rest of the group and eventually included with the Neural Crew submission.

Working on my part of the group project also gave me more practice using GitHub. I am still getting used to GitHub, so even uploading the notebook and making sure it was in the correct repository was good practice for me. I feel more comfortable doing it now than I did when we first started using GitHub. I know I still have more to learn with branches, commits, and working with other people's code, but I am starting to understand why GitHub is useful when several people are working on the same project.

Overall, I feel like I contributed by completing and checking my portion of the Module 3 lab, experimenting with the Wine model, completing the assessment and written analysis, and getting my completed notebook onto GitHub for the group. I also tried to understand what I was doing instead of just trying to finish the assignment. There were some parts that I had to slow down on, especially when looking at the code and the model results, but I think that is helping me get more comfortable with machine learning. As we continue with the class, I hope I can contribute more as I get better with Python, GitHub, and building different machine learning models.